

Nginx Web Server – Website Hosting

Question :

1. Install Nginx web server and host a simple website using the default document root. Access the website using the fake domain **firstnginx.test**.
2. Study the Nginx service, hosting types, and default configurations.

Inputs :

The following operations were performed:

- Installed **SSH** and **Nginx**.
- Started and enabled both services.
- Installed **firewalld**.
- Allowed **HTTP traffic through port 80** and reloaded the firewall.
- Modified the default **index.html** file.
- Tested the website using **curl**.
- Accessed the website using the **EC2 public IP** through Chrome.
- Added the EC2 public IP and **firstnginx.test** to the Windows hosts file.
- Accessed the website using <http://firstnginx.test>.

Commands Used :

```
dnf install sshd nginx -y
systemctl enable --now sshd
systemctl enable --now nginx
dnf install firewalld -y
systemctl enable --now firewalld
firewall-cmd --permanent --add-port=80/tcp
firewall-cmd -reload
vi /usr/share/nginx/html/index.html
```

curl localhost

Procedure

1. Installed the SSH and Nginx packages.
2. Started and enabled both services so they run automatically.
3. Installed and enabled firewalld.
4. Allowed HTTP traffic through **TCP port 80**.
5. Modified the default Nginx `index.html` page.
6. Tested the webpage locally using `curl localhost`.
7. Accessed the website through the EC2 **public IP** in Chrome.
8. Configured the Windows hosts file to map the EC2 public IP to `firstnginx.test`.
9. Accessed the website using <http://firstnginx.test> and verified that the webpage was displayed.

Output

1. Nginx Service

Output: Nginx service started successfully and was enabled.

```
[root@ip-172-31-1-172 ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-09-07 05:10:38 UTC; 1h 26min ago
  Invocation: d7c03b5e769d49f68cc444504358c603
    Main PID: 1649 (nginx)
      Tasks: 3 (limit: 4156)
     Memory: 4.2M (peak: 5.1M)
        CPU: 86ms
    CGroup: /system.slice/nginx.service
            └─1649 "nginx: master process /usr/sbin/nginx"
              └─1650 "nginx: worker process"
                └─1651 "nginx: worker process"

Sep 07 05:10:38 ip-172-31-1-172.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Sep 07 05:10:38 ip-172-31-1-172.ap-south-1.compute.internal nginx[1645]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Sep 07 05:10:38 ip-172-31-1-172.ap-south-1.compute.internal nginx[1645]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Sep 07 05:10:38 ip-172-31-1-172.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-172-31-1-172 ~]#
```

2. Website Test

The modified webpage was successfully displayed using:

curl localhost

```
[root@ip-172-31-1-172 ~]# curl localhost
<html>
<h2> Mohammed Sanjid </h2>

this is to test nginx web hosting

</html>
[root@ip-172-31-1-172 ~]# |
```

3. Access Using Public IP

The website was successfully accessed through the EC2 public IP in Chrome.



4. Access Using Fake Domain

After configuring the Windows hosts file, the website was successfully accessed using:

<http://firstnginx.test>



Result

The Nginx web server was successfully installed and configured. A simple website was hosted using the default document root and was successfully accessed through both the **EC2 public IP** and the fake domain **firstnginx.test**.

Troubleshooting

Problem

The website initially did not open when using `firstnginx.test`.

Cause

The fake-domain mapping was configured on the Linux server, but the browser was running on Windows. Windows therefore could not resolve `firstnginx.test`.

Solution

The Windows hosts file was configured to map the **EC2 public IP** to `firstnginx.test`.

After this configuration, the website was successfully accessed using the fake domain.

Study: Nginx

Nginx Service

Nginx is a high-performance web server used to serve websites and web applications. On Linux, Nginx runs as a system service and can be started, stopped, restarted, reloaded, enabled, and disabled using `systemd`.

Hosting Types in Nginx

- **Static Website Hosting** – Serves HTML, CSS, JavaScript, images, and other static files.
- **Virtual Hosting** – Allows multiple websites/domains to be hosted on the same server using different server blocks.
- **Reverse Proxy** – Nginx receives client requests and forwards them to backend applications or servers.
- **Load Balancing** – Distributes requests across multiple backend servers.
- **HTTPS Hosting** – Nginx can provide secure website access using SSL/TLS.

Default Nginx Configuration

The default website files are stored in:

`/usr/share/nginx/html/`

The default webpage is:

`/usr/share/nginx/html/index.html`

Important Nginx configuration concepts include:

- **listen** – Defines the port on which Nginx listens.
- **server_name** – Defines the hostname/domain handled by the server.
- **root** – Defines the website's document root.
- **index** – Defines the default page served to visitors.
- **Access logs** – Record client requests.
- **Error logs** – Record errors related to Nginx and requests.

Conclusion

Nginx was successfully installed and configured as a web server. A simple website was hosted using the default document root and accessed through HTTP port 80. A local fake domain, **firstnginx.test**, was configured using the Windows hosts file, demonstrating how hostname resolution can be used for local website testing.